

--	--	--	--	--	--	--	--

B.Tech. Degree VIII Semester Special Supplementary Examination September 2024

ME 19-205-0802 MATERIALS MANAGEMENT
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the scope, objectives and phases in materials management.

CO2: Get concept of static inventory problems.

CO3: Study dynamic inventory problems under risk.

CO4: Learn about lot sizing in material requirement planning.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,

L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) List out the different types of inventory problems.	3		L1	1	1
(b) Differentiate between invoice and tender.	3		L1	1	1
(c) What does a high cost of risk imply?	3		L2	2	1
(d) Explain the features of Christmas tree problems.	3		L1	2	2
(e) Explain the assumptions used in EOQ analysis.	3		L1	3	2
(f) Explain 2 bin system of inventory control.	3		L1	3	2
(g) Explain BOM file.	3		L1	4	2
(h) Explain the objectives of implementing a MRP system.	3		L2	4	2

PART B

(4 × 12 = 48)

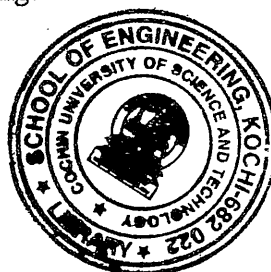
- II. The annual sales data of an automobile plant is given in the table below. Classify the items based on ABC analysis.

Items	Unit price (₹)	Number of units consumed	Items	Unit price (₹)	Number of units consumed
A	5	1000	F	1	150
B	10	10	G	8	1500
C	7	5	H	6	10000
D	750	100	I	3	20
E	5	2000	G	4	9000

OR

- III. (a) Explain vendor rating. 2 L1 1 1
- (b) Mention the factors considered in deciding the vendor rating. 4 L1 1 1
- (c) What are the different methods used to do vendor rating? 6 L1 1 1

(P.T.O.)



BTS-VIII(S/S)-09-24-2857

		Marks	BL	CO	PO										
IV.	(a) Explain the concept of EOQ. Derive the expression for EOQ.	8	L2	2	1										
	(b) Explain opportunity cost with an example.	4	L2	2	1										
OR															
V.	The manager of a flower shop promises his customers delivery within 4 hours for all orders. All flowers are purchased on a price of ₹10 per dozen and sold at ₹30 per dozen. All unsold flowers are donated to local hospital. How many dozens of flowers he need to order in order to maximize his profit? What is the optimum expected profit?	12	L3	2	1,2										
<table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <th>Dozens of flowers</th> <th>Probability</th> </tr> <tr> <td>70</td> <td>0.10</td> </tr> <tr> <td>80</td> <td>0.20</td> </tr> <tr> <td>90</td> <td>0.40</td> </tr> <tr> <td>100</td> <td>0.30</td> </tr> </table>						Dozens of flowers	Probability	70	0.10	80	0.20	90	0.40	100	0.30
Dozens of flowers	Probability														
70	0.10														
80	0.20														
90	0.40														
100	0.30														
VI.	(a) Differentiate between P systems and Q systems with neat sketches (saw tooth diagrams).	8	L1	3	1										
	(b) Explain the characteristics of dynamic inventory problems under risk.	4	L1	3	1										
OR															
VII.	(a) The demand for rivets for a shipyard is 300 numbers per year. Each rivet costs ₹36. The procurement cost is estimated as ₹30 and the interest rates are 20%. The supplier offers a discount of ₹2 per rivet for an order of 200 or above. Suggest a procurement policy considering the discount offer from the vendor.	12	L2	3	1,2										
	(b)														
VIII.	(a) Explain MRP with a block diagram. Mention the inputs and outputs of a MRP system.	8	L1	4	1										
	(b) Explain the different lot sizing methods used in MRP.	4	L1	4	1										
OR															
IX.	(a) Final product P is made from 1 unit of A and 4 units of D. For producing 1 unit of A, 3 units of C and 2 units of B are needed. The demand for product P in the 6 th , 7 th and 8 th week are 20, 10 and 50 units respectively. Determine the net requirements of item C from the following data. Lot for lot ordering strategy is followed by the company.	12	L2	4	1,2										

Items	A	B	C	D	P
Lead time(weeks)	1	2	1	2	2
Available Stock	10	20	15	50	-

Bloom's Taxonomy Levels

L1 – 45%, L2 – 35%, L3 – 20%.
